

SOP-EXCAV-800: Excavation and Trenching Safety

Document owner: Civil/Construction Department **Effective date:** 2024-03-01 **Review cycle:** 24 months

Section 1. Purpose and Scope

Governs excavation and trenching work deeper than 4 feet, including protective systems, utility locating, and atmospheric testing.

Section 2. Utility Locate

Clause 2.1 — Underground utility locating is required at least 3 working days before any excavation, and marks must be verified still visible on the day work begins or the locate is refreshed.

Clause 2.2 — Hand digging (no mechanical equipment) is required within 24 inches of any marked utility line.

Section 3. Protective Systems

Clause 3.1 — Any excavation 5 feet or deeper requires a protective system (shoring, shielding, or sloping) unless it is entirely in stable rock. Excavations 4-5 feet deep require a protective system unless a competent person determines the soil conditions present no hazard.

Clause 3.2 — Sloping angle is determined by soil classification (Type A, B, or C) per the site-specific soil analysis; a default Type C (most conservative) slope shall be used if no soil analysis has been performed.

Section 4. Competent Person Inspection

Clause 4.1 — A designated competent person shall inspect the excavation daily before entry, and after any rainfall, before allowing entry to resume.

Section 5. Atmospheric Testing

Clause 5.1 — An excavation deeper than 4 feet in an area with known or suspected contamination, or near a source of hazardous atmosphere, requires atmospheric testing before entry per the same acceptable-range criteria as SOP-CSE-200 Section 2.

Section 6. Spoil Placement and Access

Clause 6.1 — Excavated material (spoil) shall be placed at least 2 feet from the excavation edge.

Clause 6.2 — A means of egress (ladder or ramp) is required within 25 feet of travel for any excavation 4 feet or deeper.

Section 7. Records

Clause 7.1 — Daily competent-person inspection logs are filed to the Excavation Safety File and retained for 3 years.